Zhefeng Huang

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY Ph.D. in Robotics Atlanta, U.S. 08/2022-present

- Robotics Ph.D. Program, GPA: **4.00**/4.00; Passed the Robotics Ph.D. Qualifying Examination

TSINGHUA UNIVERSITY B.S. in Mechanical Engineering Beijing, China 09/2018-06/2022

- Major in Mechanical Engineering (Elite program), GPA: **3.83**/4.00; Minor in Computer Application, GPA:**3.94**/4.00

RESEARCH EXPERIENCE

World-Model-Augmented VLA for Long-Horizon Surgical Manipulation Jun 2026 – *Present*

- Developing world-model-augmented VLA policies for long-horizon surgical manipulation.
- Learning future-predictive latent states from mixed-quality expert and non-expert interactions.

Flow-Matching World Model for Robot Manipulation Failure Detection Feb 2026 – Jul 2026

- Developed an action-conditioned flow-matching world model for failure detection using only successful training data.
- Built simulation and real-world dVRK pipelines for data collection, imitation learning, and evaluation.

Self-Reconfiguration Planning for Continuum Modular Robot System Apr 2025 – Jan 2026

- Developed a mathematical model of planar continuum modular robots using combinatorial topology.
- Proposed HEART-MCTS for task planning and integrated Atlas-RRT* for motion planning.
- Developed a graph neural network (GNN)-based representation of modular robot morphology.

Mobile Nursing Robot in Intensive Care Unit Apr 2024 – Oct 2024

- Conducted a human-robot-interaction (HRI) study using a Stretch 2 robot deployed in an ICU ward.

Ultrasound-Guided Robotic Needle Insertion Sep 2023 – Jan 2024

- Developed a robotic needle insertion system using a Franka Research 3 arm with ultrasound guidance.

Development of an MR-Conditional Medical Robot Jan 2023 – Jan 2024

- Designed, developed and manufactured an MR-conditional robot prototype along with its mechatronics control system.

Geometrical Modeling and Optimization of Concentric Tubes Sep 2022 – Nov 2023

- Performed geometric modeling and optimization of concentric-tube robots for intracerebral hemorrhage evacuation.

SKILLS

Robot Learning & AI: Imitation Learning, World Models, Vision-Language-Action Models, Flow Matching
Robot Planning: Task and Motion Planning, MCTS, Sampling-based Motion Planning, Robot Kinematics, Manipulation
Programming & Frameworks: Python, C++, C, MATLAB; PyTorch, ROS 2, Isaac Sim, OpenCV, MuJoCo, Unity
Robot & Engineering Experience: da Vinci Research Kit, Franka Research 3, Stretch 2; SolidWorks, ANSYS, COMSOL

PUBLICATIONS

- **Huang, Zhefeng**, Yilin Cai, Ankit Patel, Mohammad Hajiha, Brendan Browne, and Yue Chen. “Failure Detection for Surgical Robot Imitation Policies via Flow-Matching World Modeling.” *arXiv preprint arXiv:2607.27511*, 2026. Submitted to IEEE Robotics and Automation Letters (RA-L).
- Cai, Yilin*, **Zhefeng Huang***, Yifan Wang, Haokai Xu, and Yue Chen. “Evolutionary Diversification via Modular Compliance for Self-Reconfigurable Continuum Robots.” *Science Advances* 12, eaeg9191, 2026. *These authors contributed equally to this work.
- **Huang, Zhefeng**, Anthony L. Gunderman, Samuel E. Wilcox, Saikat Sengupta, Jay Shah, Aiming Lu, David Woodrum, and Yue Chen. "Body-mounted MR-conditional robot for minimally invasive liver intervention." *Annals of Biomedical Engineering* 52, no. 8 (2024): 2065-2075.
- **Huang, Zhefeng**, Hussain Alkhars, Anthony Gunderman, Dimitri Sigounas, Kevin Cleary, and Yue Chen. "Optimal concentric tube robot design for safe intracerebral hemorrhage removal." *Journal of mechanisms and robotics* 16, no. 8 (2024).
- Gunderman, Anthony L., Saikat Sengupta, **Zhefeng Huang**, Dimitri Sigounas, Chima Oluigbo, Isuru S. Godage, Kevin Cleary, and Yue Chen. "Towards mr-guided robotic intracerebral hemorrhage evacuation: Aiming device design and ex vivo ovine head trial." *IEEE Transactions on Medical Robotics and Bionics* (2024).
- Wilcox, Samuel, **Zhefeng Huang**, Jay Shah, Xiaofeng Yang, and Yue Chen. "Respiration-Induced Organ Motion Compensation: A Review." *Annals of Biomedical Engineering* 53, no. 2 (2025): 271-283.
- Sommer, Joseph, Anthony L. Gunderman, Saikat Sengupta, **Zhefeng Huang**, Dimitri Sigounas, Kevin Cleary, and Yue Chen. "Concentric Tube Robot-Based Intracerebral Hemorrhage Evacuation in ex vivo Sheep Head: A Comparative Study." In *2024 International Symposium on Medical Robotics (ISMR)*, pp. 1-7. IEEE, 2024.

AWARDS & HONORS

Outstanding Graduate, Tsinghua University 06/2022
Three-Time Men’s Group A High Jump Champion, 62nd–64th Ma John Cup, Tsinghua University 2019-2021